IT PROF EL 4 - Advanced System Integration and Architecture

Semester Project, System Integration Documentation

INTEGRATING APPLICATIONS AND INTEGRATING THE ENTERPRISE

Submission Form

Partner 1: Florig, Abegail D.	**Partner 2: Rober, Niña Mae L.**
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Chosen App or Organization (semester project):	**GCASH**

Part 1. Integration Map and Legacy Check

1A. Integration map (list at least 4 connections between systems)

From (system)	To (system)	Data that moves	How it moves today (direct link, file, API, or by hand)
GCash Transaction Processing System	GCash Fraud Detection System	Transaction ID, user ID, amount, transaction type, timestamp	API
GCash Fraud Detection System	Customer Account System	User ID, account status, fraud risk status	API
GCash Fraud Detection System	Customer Notification Service	User ID, alert message, transaction details, alert reason	API
GCash Fraud Detection System	Fraud Monitoring Dashboard	Suspicious transaction details, risk score, alert status	API
GCash Authentication/Device System	GCash Fraud Detection System	User ID, device ID, login attempts, authentication activity, login timestamp	API

Which connections are point-to-point, and why can this become hard to manage as more systems are added? (2 to 3 sentences)

The connections from the GCash Transaction Processing System to the Fraud Detection System and from the Authentication/Device System to the Fraud Detection System are point-to-point because each system communicates directly with the fraud detection service through an API. These

connections can become difficult to manage as more systems are added because each new connection requires separate integration, security controls, monitoring, and maintenance.

1B. Legacy check

Field	Entry
Older system, tool, or process the app still depends on	Existing transaction processing and transaction-record management systems that handle payment transactions.
Why it is still used	These systems support essential transaction operations and maintain existing transaction records, so replacing them immediately could be expensive and risky.
Main challenge in connecting it	The existing transaction processing system may use a different data structure or interface from the new Fraud Detection System. The integration must correctly map transaction IDs, user IDs, amounts, and timestamps without delaying transactions or interrupting existing payment operations.
Approach to connect it (adapter or wrapper, middleware, staged replacement, data synchronization, or encapsulation)	Adapter or wrapper
Why this approach fits	An adapter can convert the existing transaction data into the format required by the Fraud Detection System while allowing the legacy system to continue operating.

Part 2. Enterprise Integration: Shared Data

Shared data item (one data item used by two or more systems):	Suspicious Transaction Record

Shared data format (at least 5 fields)

Field name	Meaning	Example value
transactionId	Unique identifier of the transaction	GC202610070001
userId	Unique identifier of the customer	USR100245
amount	Amount involved in the transaction	50000.00
riskScore	Risk score assigned by the Fraud Detection System	92
alertStatus	Current status of the fraud alert	ALERT_SENT

Systems that will use this shared format:	GCash Transaction Processing System GCash Fraud Detection System Customer Account System Customer Notification Service Fraud Monitoring Dashboard GCash Authentication/Device System

Why is one shared format better than each system using its own format? (2 to 3 sentences)

A shared Suspicious Transaction Record allows the GCash Fraud Detection System, Customer Notification Service, and Fraud Monitoring Dashboard to use the same fields, such as `transactionId`, `userId`, `riskScore`, and `alertStatus`. This reduces data conversion errors and helps ensure that the correct suspicious transaction is detected, reported, and monitored consistently across connected systems.

Part 3. Integration Strategy Choice

Field	Entry
Chosen strategy (point-to-point, API-led or service-oriented, message or event-driven, or middleware/ESB)	API-led or service-oriented
One advantage for this app	It allows the fraud detection system to securely receive transaction information and communicate with notification, authentication, monitoring, and customer-support systems through standardized services.
One disadvantage for this app	The APIs require continuous security monitoring, authentication, maintenance, version management, and availability to support real-time fraud detection.
Why this strategy fits this app (2 to 3 sentences)	The proposed fraud detection system needs to communicate with several existing GCash systems in real time. An API-led approach provides controlled and standardized interfaces for exchanging transaction and alert information without requiring every system to directly access another system's database.
One strategy not chosen, and why	Point-to-point was not chosen because the fraud detection system needs to interact with multiple systems. Many direct connections would increase complexity and make the system

	harder to maintain and secure as additional services are introduced.

Part 4. Service Entry and Best Practices

4A. Service entry (one specific service, not the whole app)

Field	Entry
Service name	Fraud Alert Service
What it does	Receives suspicious-transaction information from the fraud detection system and sends a real-time alert to the affected customer and authorized personnel when necessary.
Who uses it	GCash Fraud Detection System, authorized fraud-monitoring personnel, and the customer notification system.
Input (what it needs)	Transaction ID, user ID, transaction amount, risk score, transaction type, and alert reason.
Output (what it returns)	Alert ID, transaction ID, alert status, and timestamp
Data format (XML or JSON) and why	JSON – JSON is lightweight, easy for applications to process, and suitable for real-time API communication between web, mobile, and backend systems.
Who is responsible for it	GCash IT, cybersecurity, and fraud-management teams
Security control (for example, an API key)	OAuth 2.0 authentication/access token, HTTPS/TLS encryption, and role-based authorization
Version (start at v1)	v1

4B. Best practices applied (2 XML or data format practices and 2 web service practices)

Type	Practice chosen	How it applies to this service (one sentence)
XML or data format practice 1	Consistent data structure	The Fraud Alert Service uses the same `transactionId`, `userId`, `riskScore`, and `alertStatus` field names in every alert request and response.
XML or data format practice 2	Data validation	Before creating an alert, the Fraud Alert Service checks that `transactionId` and `userId` are present, `amount` is a valid non-negative number, and `riskScore` is within the allowed range of 0–100.
Web service practice 1	Secure communication	The Fraud Alert Service uses HTTPS/TLS to protect the `userId`,

		`transactionId`, and alert details while transmitting them to the Customer Notification Service.
Web service practice 2	Authentication and authorization	The Fraud Alert Service verifies the application's access token and checks whether the requesting Fraud Detection System or authorized fraud-monitoring personnel have permission to create or view alerts.